BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, DI KHAN

SECONDARY SCHOOL (9TH CLASS) EXAMINATION, 2021

CHEMISTRY

Past Paper 2021 • Paper Code: DIK-SSC-IX-2021(A)

Retyped faithfully from the original board paper — same questions, no changes.

SECTION – A

Time Allowed: 20 min | **Maximum Marks:** 12 | **Paper Code:** DIK-SSC-IX-2021(A)

Q1. Choose the correct answer from four possible answers (A,B,C,D) and fill the correct answer on MCQ's sheet separately.

Q1. An atom with a charge is called_____

(A) An electron
(B) A molecule
(C) A metal
(D) An ion

Q2. Hydrochloric acid, nitric acid and white lead prepared by:

(A) Al-Razi
(B) Jabber Ibn Haiyan
(C) Ibn-e-Sina
(D) Al-Beruni

Q3. Which of the following would have the largest ionization energy?

(A) Na
(B) Al
(C) H
(D) He

Q4. Which statement for the particles of solid is not correct?

(A) They move at greater speed
(B) They are arranged in regular pattern
(C) There is a very little space between the particles
(D) The force of attraction between the particles are strong

Q5. In which of the following does sulphur have an oxidation number of +7?

(A) HSO_3
(B) SO_3
(C) H_2SO_4
(D) $H_2S_2O_8$

Q6. Electron in its ground state does not _____

(A) Spin
(B) Revolve
(C) Radiate energy
(D) Reside in orbit

Q7. 2 moles of Na_2SO_4 are dissolved in one dm^3 of solution, molarity of solution is _____

(A) 1M (B) 2M

(C) 3M (D) 0.5M

Q8. Halogens react with non-metals forming respective_____

(A) Ionic halides (B) Covalent halides

(C) Both A and B (D) None of these

Q9. All the ionic compounds are represented by _____

(A) Molecular formula (B) Empirical formula

(C) Formula unit (D) Chemical formula

Q10. The bond in CO_2 is _____

(A) Single covalent bond (B) Double covalent bond

(C) Triple covalent bond (D) Ionic bond

Q11. Group III to group VIII elements are called _____elements.

(A) s-block (B) d-block

(C) p-block (D) f-block

Q12. Which of the following is weak electrolytic?

(A) HCl (B) NaCl

(C) KCl (D) NH_4OH

SECTION – B

Time Allowed: 2:40 Hrs | **Maximum Marks:** [illegible in source scan] | **Paper Code:** DIK-SSC-IX-2021(A)

Q2. Attempt any 8 questions of the following:

(i) Why do atoms forms chemical bonds?

(ii) Define element and substance. Write the symbol of four different elements.

(iii) Place the following element in order of increasing ionization energy. Na, S, Mg and Ar.

(iv) Why the vapour pressure of a liquid increase with increase in temperature? Give reason.

(v) Define electrolytes. What is the difference between strong and weak electrolytes?

(vi) What volume of water must be added to $85cm^3$ of 3.5M Na_2CO_3 to dilute it to 0.41M?

(vii) Differentiate between atom and ion. Give example.

(viii) Identify at least two groups which contain metallic elements.

(ix) Write the electronic configuration of $_{7}N$ and $_{9}F$.

(x) What is molarity? Write its formula.

(xi) How do the first ionization energy of representative element vary across a period and down a group?

SECTION – C

Maximum Marks: 24

> Note: Attempt any 3 questions: Each carry 8 marks.

Q3.

(a) Explain covalent bond on the basis of one example.

(b) Make a general sketch of the periodic table showing s,p,d and f block element (without showing the symbol of elements).

Q4.

(a) Why ethyl alcohol is higher vapour pressure then water? Give reason.

(b) Summarize the fundamental particles of an atom.

Q5.

(a) What were the principal goals of Muslim chemist? Why the Muslim period is generally called "Al-Chemist" in the history of chemistry?

(b) Differentiate between saturated and unsaturated solution. Give example.

Q6.

(a) Sketch the "Deinel Cell". Labelling the cathode, anode and the direction flow of electrons.

(b) Compare the properties of metals and non-metals.